

Mikael Yeghiazaryan

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EDUCATION

PhD in Computer Science, University of Illinois Urbana-Champaign	Champaign, IL, USA <i>August 2026 – present</i>
Master of Engineering Science (MEng), University of Oxford	Oxford, UK <i>June 2021 – June 2022</i>
• Graduated with First Class Honours; ranked 15th out of 159. • Awarded the Swire Scholarship for outstanding academic achievement.	
Bachelor of Engineering Science (BEng), University of Oxford	Oxford, UK <i>October 2018 – June 2021</i>

WORK EXPERIENCE

Researcher <i>Center for Scientific Innovation and Education</i> • Led a research project on verifiably safe autonomous systems. • Lectured a course on Computer Vision.	Yerevan, Armenia <i>September 2025 – August 2026</i>
Research Assistant <i>Advanced Controls Research Laboratory, University of Illinois Urbana-Champaign</i> • Supervised by Dr. Naira Hovakimyan.	Urbana-Champaign, IL, USA <i>July 2024 – June 2025</i>
Research Assistant <i>Robotics Institute, Carnegie Mellon University</i> • Supervised by Dr. Fernando De La Torre and Dr. Jessica Hodgins.	Pittsburgh, PA, USA <i>September 2022 – July 2024</i>
Research Intern <i>Oxford Robotics Institute, University of Oxford</i> • Supervised by Dr. Maurice Fallon (ORI).	Oxford, UK <i>June 2021 – August 2021</i>

PUBLICATIONS & RESEARCH PROJECTS

The 37th IEEE International Symposium on Software Reliability Engineering • Title: Synergistic Simplex: Cooperative Runtime Assurance for Safety-Critical Autonomous Systems • Authors: Ayoosh Bansal*, Mikael Yeghiazaryan*, Artyom Khachatryan, Tianyi Zhu, Hunmin Kim, Naira Hovakimyan, Lui Sha. (* – Equal contribution) • Designed a cooperative runtime assurance architecture that incorporates machine learning information into the safety layer, improving performance while preserving formal safety guarantees of autonomous systems.	October 2026
IEEE International Conference on Image Processing (ICIP) 2025 • Title: <i>Texture- and Shape-based Adversarial Attacks for Overhead Image Vehicle Detection.</i> • Authors: Mikael Yeghiazaryan, Sai Abhishek Siddhartha Namburu, Emily Kim, Stanislav Panev, Celso de Melo, Fernando De la Torre, Jessica Hodgins. • Developed and evaluated physically realizable adversarial attack strategies for aerial vehicle detectors, balancing attack effectiveness with real-world texture and shape constraints.	September 2025
AIAA Aviation Forum and Exposition 2025 • Title: <i>AirTaxiSim: A Simulator for Autonomous Air Taxis.</i> • Authors: Ayoosh Bansal*, Mikael Yeghiazaryan*, Hyung-Jin Yoon*, Duo Wang, Petros Voulgaris, Naira Hovakimyan, Lui Sha. (* – equal contribution) • Developed <i>AirTaxiSim</i>, a high-fidelity simulation framework for evaluating and benchmarking autonomous air-taxi operations in complex urban environments.	July 2025
AIAA SciTech Forum and Exposition 2025 • Title: <i>Verification and Validation of a Vision-Based Landing System for Autonomous VTOL Air Taxis.</i>	January 2025

- Authors: Ayoosh Bansal*, Duo Wang*, Mikael Yeghiazaryan*, Yangge Li, Chuyuan Tao, Hyung-Jin Yoon, Prateek Arora, Christos Papachristos, Petros Voulgaris, Sayan Mitra, Lui Sha, Naira Hovakimyan. (* – equal contribution)
- Developed a formal verification and validation framework leveraging high-fidelity simulation to evaluate the safety of a vision-based landing system for autonomous VTOL air taxis operating in cluttered urban environments.

Augmenting Aerial Imagery using Vision-Language Models | CMU September 2023 – December 2024

- Supervision: Dr. Fernando De la Torre and Dr. Jessica Hodgins.
- Developing methods to enhance aerial and satellite imagery for vehicle detection by leveraging diffusion and vision-language models (e.g., Stable Diffusion). Conducting adversarial analyses of generative models to assess robustness and explain detector behavior using language.

4th year Master Thesis Project | University of Oxford September 2021 – June 2022

- Title: Learning generalizable keypoints for object pose estimation.
- Supervision: Dr. Joao Henriques and Dr. Dylan Campbell (Visual Geometry Group, University of Oxford).
- Developed a method for unseen object pose estimation using generalizable features and low-resolution CAD models.

3rd year Group Project | University of Oxford September 2020 – May 2021

- Title: Formula Student Electric Vehicle (EV) Race Car Design.
- Supervision: Dr. Dan Rogers (University of Oxford).
- Designed the electronics and control systems for a Formula Student electric race car.

HONOURS AND AWARDS

1st Class Honours Oxford, UK *University of Oxford* *October 2018 – June 2022*

- I achieved and graduated with 1st Class Honours (equivalent to 4.0 GPA) in all years of examinations at the University of Oxford.
- Awarded the Swire Scholarship for exceptional academic performance at my college (University College Oxford).

Honorable mention in The International Physics Olympiad Jogjakarta, Indonesia *International Physics Olympiad, 2017* *July 2017*

Silver medal in The International Zhautykov Olympiad (Physics) Almaty, Kazakhstan *International Zhautykov Olympiad, 2017* *January 2017*

COMMUNITY & LEADERSHIP

Secretary and President of the Oxford University Armenian Society Oxford, UK *Oxford University Armenian Society (OUAS)* *October 2020 – June 2022*

- I set up events for the society members.
- I engaged in the promotion of the society's events at the university level.

Lecturer of Engineering Science Yerevan, Armenia *"Quantum" Gymnasium* *August 2019*

- I lectured engineering of electronics at my alma mater's summer camp for high school students.

President of the Experimental Physics Club Yerevan, Armenia *"Quantum" Gymnasium* *October 2016 – March 2018*

- I founded a club for conducting experiments in physics and helped other students with their assignments.

SKILLS

Programming: Python, C/C++, MATLAB.

Machine Learning: PyTorch, Transformers, CLIP, Diffusion Models, TensorFlow, Keras, scikit-learn, pandas.

Computer Vision & Graphics: OpenCV, NumPy, PyTorch3D, Kaolin, Blender, CARLA, PyBullet.

Systems & Tools: Linux (Ubuntu), Git, ROS, Docker, Anaconda, Flask, AWS, PostgreSQL, Arduino, Raspberry Pi, L^AT_EX.

Research: Critical thinking, teamwork, technical communication, presentation, and clean coding.

MISCELLANEOUS

Languages: Armenian (native), Russian (native), Belarusian (native), English (fluent).

Music: Violinist since age 6; laureate of international competitions; performed at Vladimir Spivakov's festival at age 10.

Chess: Achieved Class A Elo ranking at age 12, now playing recreationally (Lichess ID Michael.Yeghiazaryan).